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ADDITIONAL MOTOR MANUAL

# **Technical manual for low voltage IP23 IC01 electric motors M3FH 280-355**



# Contents

**Validity ..... 3**

**Installation ..... 3**

**Operating conditions..... 4**

**Maintenance ..... 5**

**Safety ..... 6**

**Contact us ..... 7**

## Validity

This additional installation, operation, and maintenance manual is valid for an air-cooled motors with IC01 cooling classification (M3FH 280–355) and is to be read together with the main manual (Low Voltage Motors/Manual). Additional information may be required for some machines due to special application and/or design considerations.

## Installation

The cooling air intake for the motor is at the opposite end of the terminal box location (see Figure 1). Ensure free flow of cooling air and control the correct air flow direction. Also lead away the hot cooling air of other potential heat sources. The ventilator is designed only to cool the motor itself, not other equipment on the installation location.

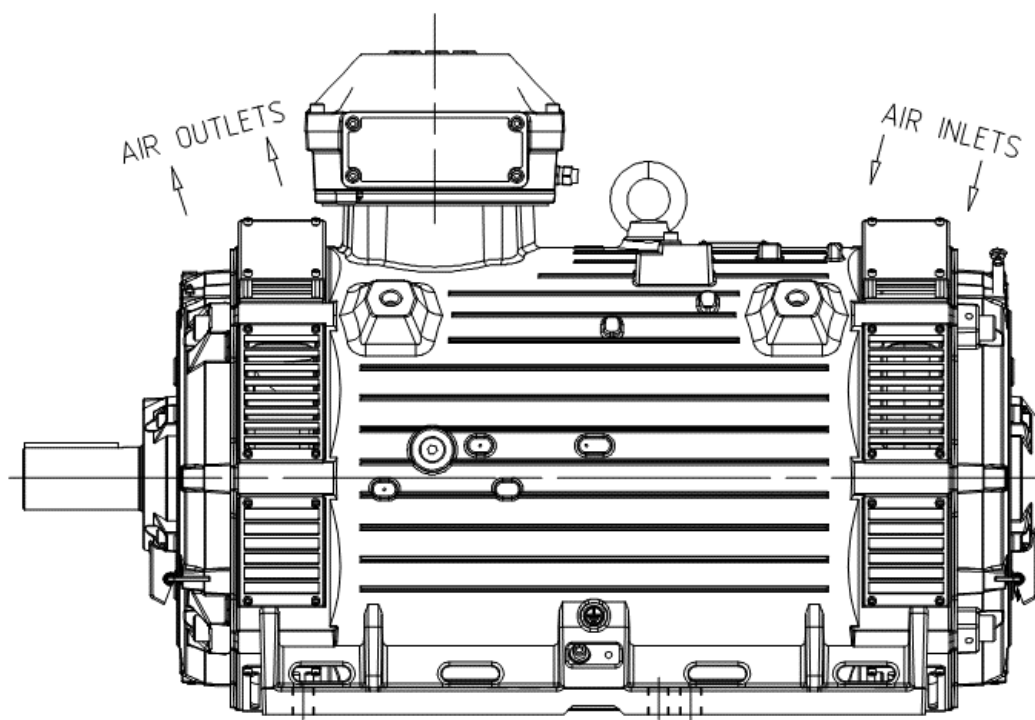


Figure 1. Location of air inlets / outlets, and airflow in the motor.

Normal ambient temperature range is from  $-20\text{ }^{\circ}\text{C}$  to  $+40\text{ }^{\circ}\text{C}$  for cooling air intake. Deviation to this range is separately agreed. Also rating plate defines deviations to normal temperature range. With M3FH cooling principle the intake air temperature will be 5–10°C higher than ambient overall ambient temperature. Lead, if possible, the hot exhaust air away from coolant intake.

Check that the motor has sufficient cooling for installation location. Ensure that no nearby objects or direct sunshine radiate additional heat to the motor. The motor should also be mounted on a flat and rigid surface. Vibration levels must not exceed ISO

20816-3 Zone B/C (4,5 mm/s RMS) when installed and running the application. With higher vibration levels, please contact ABB.



The cooling air temperature at outlet is increased approx. 20K. Do not install cables or other material sensitive to heat in the exhaust air flow.



A fault in the motor windings could cause an arc and spread molten metal droplets out of the machine. Do not install anything inflammable near the machine. Do not stay unnecessarily close to the motor while it's running.



Make sure that anti-condensation heaters (if installed) are turned off when the motor is running.



The outlets of the motor emit hot air. To prevent burns of other related injuries, please ensure that no personnel are close to the outlet when the motor is in operation. Maintain a safe distance and use appropriate protective equipment.

## Operating conditions

Motor series M3FH is designed to be used in clean and dry environments. IP23 motors are most suitable for indoor installations where there is minimal exposure to dust, moisture and/or other contaminants. To prevent overheating, ensure that the motor is installed in a location that has with proper ventilation. Also, it is very important to ensure that the ventilation around the motor is sufficient, so that the recirculation of warm air back to the system is prevented. These motors should not be used in conditions where there are high level of contaminants. Direct water exposure to the motor must be prevented. If the motor is to be used outdoors, ensure that the motor and the surrounding area is properly protected to prevent dust and moisture from entering the motor.

Consider using additional protective measures such as protective filters that are designed to prevent large particles from entering the motor, if the motor is going to be installed in a environment with harsh conditions. If the motor is equipped with inlet air filters, periodic maintenance and inspection should be performed. Change or wash the protective filters when necessary. Motor must be operated in a specified temperature range of of -20°C to +40°C. If the motor is exposed to high humidity and/or temperature fluctuations, use anti-condensation heaters.



Do not wash the filters when the motor is running!



The velocity of the cooling air is high. Protect the eyes when staying near motors whilst running. The standard M3FH motor have no air intake filters. The motor must not be installed outdoors. The machines are not protected against ingress of e.g. snow, leaves, metal chips, oil mist etc. The machines must not be used in the presence of dust of organic origin or other kind of flammable dust because such an environment is hazardous.



The motor has a high noise level. Protect the ears!



Allowed starting time for M3FH motors must be double to starting time for TEFC-types (e.g. M3BP).

## Maintenance

Due to the open design of IP23 motors, regular maintenance activities are crucial. Inspect the motor for dust and contaminant accumulation, and clean the cooling ducts frequently. Ventilation air passages must not be blocked in any scenario. Check for any signs of corrosion or moisture ingress and take corrective actions as needed. If the motor is equipped with anti condensation heaters, make sure they are functioning properly to help prevent moisture buildup.

Make sure that the ventilation outlets and inlets are not blocked by any kind of debris. Outlets and inlets need to be clear of dust and other contaminants to ensure proper airflow and prevent overheating.

Follow the manufacturer's recommendations for lubrication intervals and types of lubricants. Regularly check and replenish lubrication to prevent bearing wear.

Frequently inspect the motor's electrical connections and insulation resistance. Look for signs of wear or damage and take corrective actions as needed.



All maintenance tasks, except for greasing, should be performed only when the machine is in non-operating condition.



Dust and other contamination buildup in the motor ventilation paths, such as grills and cooling ribs, will lead the motor to operate in a higher temperature than it normally should. This might eventually decrease the winding lifespan, and cause premature thermal failure.



Contaminants will also erode the winding, which might degrade the winding and compromise the insulation, and therefore potentially leading to a electrical failure, such as short circuits and ground faults. Dust can also cause a conductive path between winding parts, possibly leading to a short circuit.

## Safety

If the motor's ventilation covers are removed for dust inspection, the motor must be disconnected from the electrical network and ensured that it cannot be reconnected. Additionally, several other safety measures must be taken:

- Appropriate personal protective equipment, such as gloves, safety glasses, and a breathing mask must be used
- Adequate ventilation in the area should be ensured to avoid inhaling dust particles
- The electric motor must be allowed to cool down to avoid burns



Confirm that the motor is electrically and mechanically disconnected from the network before dust inspection and maintenance !



## Contact us

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